Non-Linear Regression Analysis

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Introduction

In this project, a given dataset of paired observations $(t, y(t))$ is analyzed using three proposed non-linear models. The objective is to determine the model that best represents the data by estimating the unknown parameters through the least squares method.

1 Least Squares Estimators of the Parameters

Models and Estimated Parameters

Model	Estimated Parameters
Model 1: $y(t) = \alpha_0 + \alpha_1 e^{\beta_1 t} + \alpha_2 e^{\beta_2 t}$	$\hat{\alpha}_0 = 0.567060$, $\hat{\alpha}_1 = 1.099952$, $\hat{\alpha}_2 = -0.689433$, $\hat{\beta}_1 = -0.179741$, $\hat{\beta}_2 = -0.867731$.
Model 2: $y(t) = \dfrac{\alpha_0 + \alpha_1 t}{\beta_0 + \beta_1 t}$	$\hat{\alpha}_0 = -16.401511$, $\hat{\alpha}_1 = -6.417100$, $\hat{\beta}_0 = -72.974447$, $\hat{\beta}_1 = -10.218347$.
Model 3: $y(t) = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3 + \beta_4 t^4$	$\hat{\beta}_0 = 0.301192$, $\hat{\beta}_1 = 0.000209$, $\hat{\beta}_2 = 0.001812$, $\hat{\beta}_3 = -0.000069$, $\hat{\beta}_4 = 0.000001$.

2 Method of Estimation and Initial Guesses

2.1 Estimation method for Model 1 (Newton Method)

Model 1 is

$$y(t) = \alpha_0 + \alpha_1 e^{\beta_1 t} + \alpha_2 e^{\beta_2 t} + \varepsilon(t),$$

which is linear in $\alpha = (\alpha_0, \alpha_1, \alpha_2)^\top$ and nonlinear in $\beta = (\beta_1, \beta_2)^\top$. Define the $n \times 3$ design matrix (parametrized by β)

$$X(\beta) = \begin{bmatrix} 1 & e^{\beta_1 t_1} & e^{\beta_2 t_1} \\ 1 & e^{\beta_1 t_2} & e^{\beta_2 t_2} \\ \vdots & \vdots & \vdots \\ 1 & e^{\beta_1 t_n} & e^{\beta_2 t_n} \end{bmatrix}.$$

For any fixed β the optimal linear coefficients have the closed form

$$\hat{\alpha}(\beta) = \big(X(\beta)^\top X(\beta)\big)^{-1} X(\beta)^\top y.$$

Write the residual sum of squares as

$$\mathrm{SSE}(\beta) = \|y - X(\beta)\hat{\alpha}(\beta)\|^2 = y^\top y - Q(\beta), \qquad Q(\beta) \; = \; y^\top X(\beta)\big(X(\beta)^\top X(\beta)\big)^{-1} X(\beta)^\top y.$$

Minimizing $\mathrm{SSE}(\beta)$ is equivalent to *maximizing* $Q(\beta)$; hence we solve $\min_\beta -Q(\beta)$.

Let the gradient and Hessian of the objective function be denoted as

$$G = \begin{bmatrix} \dfrac{\partial Q}{\partial \beta_1} & \dfrac{\partial Q}{\partial \beta_2} \end{bmatrix}, \qquad H = \begin{bmatrix} \dfrac{\partial^2 Q}{\partial \beta_1^2} & \dfrac{\partial^2 Q}{\partial \beta_1 \partial \beta_2} \\ \dfrac{\partial^2 Q}{\partial \beta_2 \partial \beta_1} & \dfrac{\partial^2 Q}{\partial \beta_2^2} \end{bmatrix}.$$

Then, the Newton iteration is given by

$$\beta^{(a+1)} = \beta^{(a)} - \lambda\, H^{-1} G,$$

where $0 < \lambda \leq 1$ is the step factor used for damping or backtracking to ensure convergence.

Initialisation. The initial guesses for (β_1, β_2) are chosen by a grid search over

$$\mathcal{G} = \{-1,\, -0.5,\, -0.2,\, 0.2,\, 0.5,\, 1\}^2,$$

excluding pairs with $|\beta_1 - \beta_2| < 10^{-3}$ to avoid near-collinearity. The pair from $\mathcal{G}$ yielding the largest $Q(\beta)$ (or smallest SSE) is used as the starting point $\beta^{(0)}$.

After convergence of β we compute $\hat{\alpha} = \hat{\alpha}(\beta)$ and report $(\hat{\alpha}, \hat{\beta})$.

2.2 Estimation method for Model 2 (Newton Method)

For Model 2,

$$y(t) = \frac{\alpha_0 + \alpha_1 t}{\beta_0 + \beta_1 t} + \varepsilon(t),$$

the design matrix is defined as

$$X(\beta) = \begin{bmatrix} \dfrac{1}{\beta_0 + \beta_1 t_1} & \dfrac{t_1}{\beta_0 + \beta_1 t_1} \\ \vdots & \vdots \\ \dfrac{1}{\beta_0 + \beta_1 t_n} & \dfrac{t_n}{\beta_0 + \beta_1 t_n} \end{bmatrix}.$$

The gradient and Hessian of the objective function are denoted by

$$G = \begin{bmatrix} \dfrac{\partial Q}{\partial \beta_0} & \dfrac{\partial Q}{\partial \beta_1} \end{bmatrix}, \qquad H = \begin{bmatrix} \dfrac{\partial^2 Q}{\partial \beta_0^2} & \dfrac{\partial^2 Q}{\partial \beta_0 \partial \beta_1} \\ \dfrac{\partial^2 Q}{\partial \beta_1 \partial \beta_0} & \dfrac{\partial^2 Q}{\partial \beta_1^2} \end{bmatrix}.$$

The iteration step and grid-search algorithm remain the same as done in Model 1.

2.3 Estimation method for Model 3

For Model 3, the model is linear in all parameters; hence, the least squares estimators are obtained directly using the normal equations:

$$\boxed{\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}.}$$

No initial guesses are required for this model.

3 Selection of the Best Fitted Model

Model	Model Equation	SSE	R^2 Score
Model 1	$y(t) = \alpha_0 + \alpha_1 e^{\beta_1 t} + \alpha_2 e^{\beta_2 t}$	2.083801	0.162764
Model 2	$y(t) = \dfrac{\alpha_0 + \alpha_1 t}{\beta_0 + \beta_1 t}$	2.204509	0.114266
Model 3	$y(t) = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3 + \beta_4 t^4$	2.0096854	0.192543

Table 1: Comparison of Model Equations, SSE, and R^2 Scores

As the Sum of Squared Errors (SSE) for **Model 3** is the smallest ($SSE_3 < SSE_1 < SSE_2$), it indicates that **Model 3 provides the best fit to the data**. Also the R2 score for model 3 is the best. **Hence, all subsequent analysis is done on Model 3.**

4 Estimation of Error Variance

For the selected Model 3,

$$y(t) = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3 + \beta_4 t^4 + \varepsilon(t),$$

Let $\hat{y} = X\hat{\beta}$ denote the fitted values and $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ the residual vector. The estimator of the error variance σ^2 is given by

$$\hat{\sigma}^2 = \frac{\mathbf{e}^\top \mathbf{e}}{n - p},$$

where n is the number of observations and p is the number of estimated parameters.
For Model 3, $n = 55$ and $p = 5$, hence

$$\hat{\sigma}^2 = \frac{2.0096854015858603}{55 - 5} = 0.04019370803171721.$$

Therefore, the estimated value of the error variance is

$$\boxed{\hat{\sigma}^2 = 0.04019370803171721}.$$

5 Confidence intervals based on the Fisher information matrix.

The Fisher Information Matrix for the normal linear model is derived from the log-likelihood as follows:

$$I(\beta) = E\left[-\frac{\partial^2 \ell(\beta)}{\partial\beta\,\partial\beta^\top}\right]$$

$$f(y; \beta, \sigma^2) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta)\right)$$

$$\ell(\beta, \sigma^2) = \log f(y; \beta, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta)$$

$$\boxed{I(\beta) = \frac{1}{\sigma^2} X^\top X}$$

Consider σ^2 to be the true variance (unknown), we cannot construct a confidence interval using equation (1) given below:

$$\mathrm{Var}(\hat{\beta}) = I(\beta)^{-1} = \sigma^2 (X^\top X)^{-1}$$

$$y_i = x_i^\top \beta + \varepsilon_i, \quad i = 1, 2, \ldots, n$$
$$y = X\beta + \varepsilon$$
$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$
$$\hat{\beta} - \beta = (X^\top X)^{-1} X^\top \varepsilon$$

$$\boxed{Z = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\sigma^2 \, [(X^\top X)^{-1}]_{jj}}} \sim N(0, 1)} \tag{1}$$

Hence we need to find the distribution of

$$\boxed{\frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2 \, [(X^\top X)^{-1}]_{jj}}}} \tag{2}$$

to find the confidence interval of β_j.
Let

$$W = \frac{(n-p)\,\hat{\sigma}^2}{\sigma^2}, \quad \hat{\sigma}^2 = \frac{\text{SSE}}{n-p} = \frac{\varepsilon^\top (I - H)\varepsilon}{n-p}, \quad H = X(X^\top X)^{-1} X^\top.$$

Here, $I - H$ is **symmetric** and **idempotent**. Given that $\varepsilon \sim N_n(0, \sigma^2 I_n)$, define $z = \frac{\varepsilon}{\sigma} \Rightarrow z \sim N_n(0, I_n)$.

$$W = z^\top (I - H) z.$$

Since H projects onto the column space of X (rank p), $I - H$ projects onto its orthogonal complement of dimension $n - p$. As $I - H$ is symmetric,

$$I - H = P^\top \Lambda P,$$

where P is orthogonal and $\Lambda = \text{diag}(1, \ldots, 1, 0, \ldots, 0)$ with $n - p$ ones and p zeros.

$$W = z^\top P^\top \Lambda P z = (Pz)^\top \Lambda (Pz) = \sum_{i=1}^{n-p} (Pz)_i^2.$$

Since $Pz \sim N_n(0, I_n)$, we obtain

$$\boxed{W \sim \chi^2_{n-p}} \tag{3}$$

From (1) & (2) , the statistic

$$\boxed{T_j = \frac{Z}{\sqrt{W/(n-p)}} \sim t_{n-p}} \tag{4}$$

Finally, the $(1-\alpha)$-level confidence interval for β_j is given by (where $\alpha = 0.05$ corresponds to a 95% confidence interval):

$$\boxed{\hat{\beta}_j \pm t_{1-\alpha/2,\, n-p} \, \hat{\sigma} \sqrt{\left[(X^\top X)^{-1}\right]_{jj}}} \tag{5}$$

Parameter	**Estimate**	**Standard Error**	**95% Confidence Interval**
β_0	0.301192	0.151356	[-0.002816, 0.605201]
β_1	0.000209	0.036731	[-0.073568, 0.073985]
β_2	0.001812	0.002634	[-0.003478, 0.007101]
β_3	-0.000069	0.000070	[-0.000211, 0.000072]
β_4	0.000001	0.000001	[-0.000001, 0.000002]

Table 1: Coefficients, standard errors and 95% confidence intervals for Model 3.

6 Plot of Residuals

For Model 3, the residuals are defined as

$$\hat{e}_i = y_i - \hat{y}_i, \qquad i = 1, 2, \ldots, n,$$

where $\hat{y}_i$ denotes the fitted value obtained from the polynomial model

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 t_i + \hat{\beta}_2 t_i^2 + \hat{\beta}_3 t_i^3 + \hat{\beta}_4 t_i^4.$$

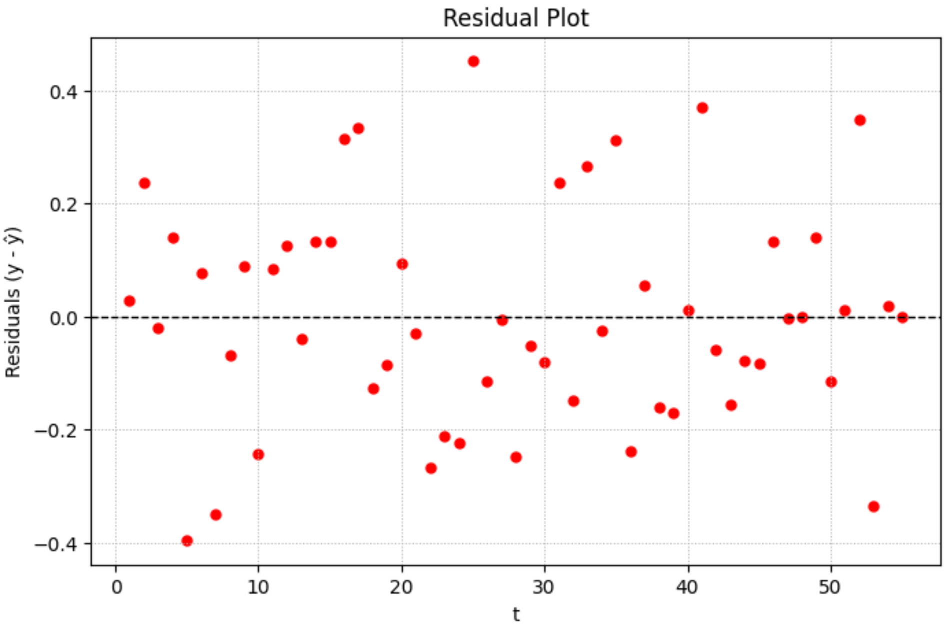

Figure 1: Plot of residues with t for Model 3.

7 Test whether it satisfies the normality assumption or not

To verify the normality assumption for the linear model 3 (best-fit), the **Shapiro–Wilk test** was conducted, and the corresponding p-value was computed for Model 3. A p-value greater than 0.05 indicates that the normality assumption holds.

For model-3: $W = 0.983220 \qquad p = 0.635375$

A higher W value (closer to 1) and a large p-value indicate that the sample distribution does not significantly deviate from normality.

Additionally I plotted a QQ Plot.
A **Q–Q plot** compares the quantiles of the sample data with those of a theoretical distribution (e.g., normal). If the data follows the distribution, the points lie approximately along a 45° line as can be seen below.

From the Shapiro–Wilk Test & the Q–Q Plot, it is evident that Model 3 follows the normality assumption.

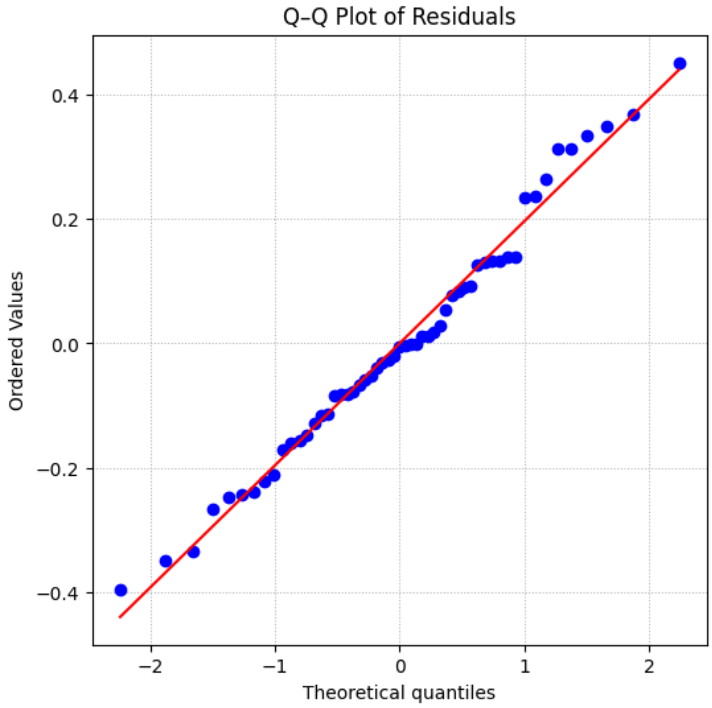

Figure 2: Q–Q Plot of Residuals for Model 3

8 Plot of Observed Data & Fitted Curve

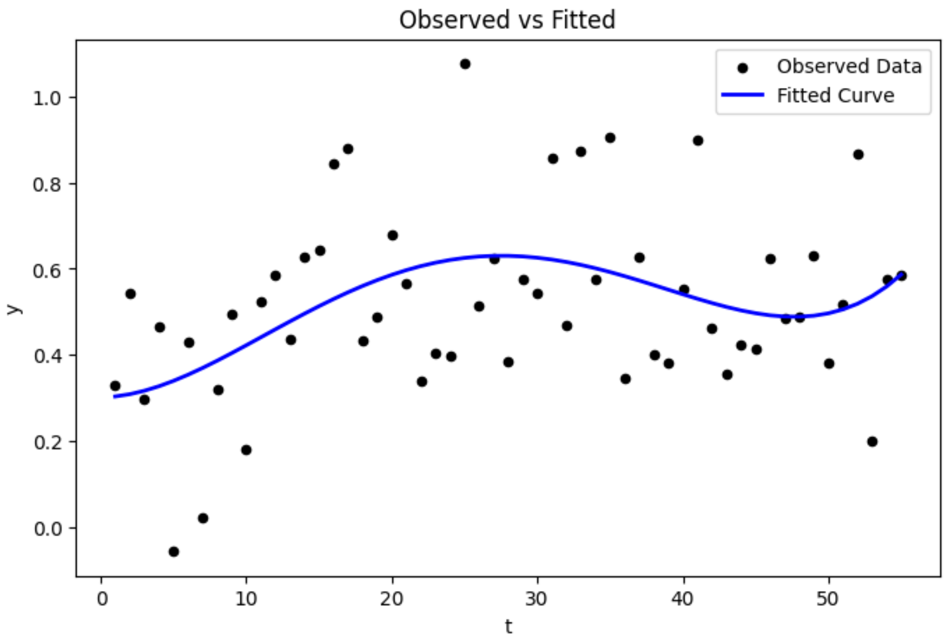

Figure 3: Plot of observed data points and the fitted curve for Model 3.